National University of Computer and Emerging Sciences, Lahore Campus

Course:	Multivariable Calculus	Course Code:	MT306
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	21-Feb-17	Weight	15%
Section:	EE-1	Page(s):	1
Exam:	Mid 1		

L15-4522
EE-2A

Question #01: [5 marks]

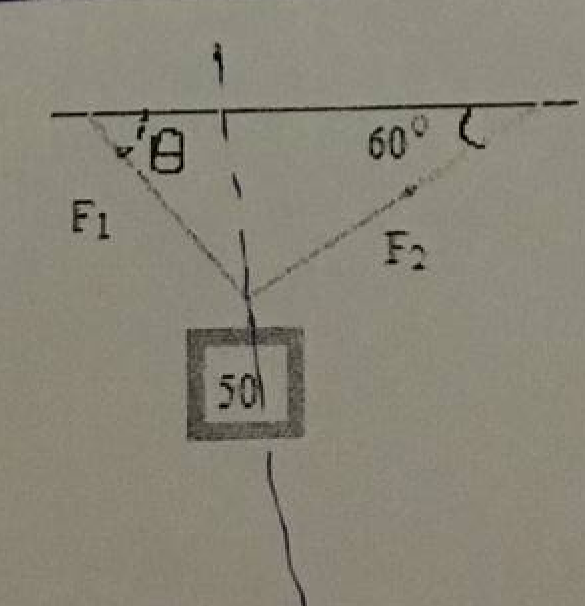
Find values of x such that the angle between the vectors $v = (2, 1, -1)$ and $w = (1, x, 0)$ is 45^0. Compute the value of x.

Question #02: [5+5 marks]
Find parametric equations for the line of intersection of the planes and also find the angle between the planes.

$$x + y + z = 1$$
$$x + 2y + 2z = 1$$

Question #03: [5 marks]
A helicopter is to fly directly from a helipad at the origin in the direction of the point (1,1,1) at a speed of $60 ft/sec$. What is the position of the helicopter after 10 sec?

Question #04: [10 marks]
Consider a 50-N weight suspended by two wires as shown in figure. If the magnitude of the vector F_1 is 35N, find the angle θ and the magnitude of vector F_2.



National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Multivariable Calculus	Course Code:	MT-306
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	180 Minutes	Total Marks:	70
Paper Date:	22-05-2017	Weight	50
Section:	ALL	Page(s):	2
Exam Type:	Final		

Student : Name:________________ Roll No.________________ Section:________

Instruction/Notes:
1. Attempt all questions. Exchange of calculators or programmable calculators are not allowed.
2. If you think something wrong or need to be modified, do it with the best of your understanding.

Question #01: [10 marks]

a) If $f(x,y) = x^2y^3 - x^3y^2 + 3x + 2y$, compute the limit of $f(x,y)$ as $(x,y) \to (1,2)$ and also find the points where the function is discontinuous, if any.

b) Use implicit differentiation to find $\frac{\partial z}{\partial x} = 0$ and $\frac{\partial z}{\partial y}$

$$yz + x\ln y = z^2$$

(i) $\frac{-\ln y}{y - 2z}$

(ii) $\frac{-x - 2y}{y(y-2z)}$

Question #02: [10 marks]

Suppose that over a certain region of space the electrical potential V is given by

$$V(x,y,z) = 5x^2 - 3xy + xyz.$$

(a) Find the rate of change of the potential at $P(3,4,5)$ in the direction of the vector $v = i + j - k$.
(b) In which direction does V change most rapidly at P?
(c) What is the maximum rate of change at P ?

Question #03: [10 marks]

a) Determine whether or not the vector field is conservative. If it is, find a function f such that $F = \nabla f$, where $F(x,y) = x^2y^3\, i + x^3y^2 j$. $3x^2y^2$ $3x^2y^2$

b) Use part (a) to find work done along the given curve C, C: $\mathbf{r}(t) = (t^3 - 2t,\ t^3 + 2t), 0 \le t \le 1$.

Question #04: [10 marks]

Find the volume of the given solid that lies under the surface $z = xy$ and above the triangle region D in $xy-$ plane with vertices $(1,1), (4,1), (1,2)$. $\frac{31}{8}$

Question #05: [10 marks]

Find the surface area of the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the plane $z = 1$.

$P\,dx + Q\,dy$ $\frac{16\pi}{3}$

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Question #06: [10 marks]

Find the work done by the force $F(x, y) = x(x + y)i + xy^2j$ in moving a particle from the origin along the x-axis to (1,0), then along the line segment to (0,1), and then back to the origin along the y-axis by using Green's theorem. $-3/4$?

Question #07: [10 marks]

Find flux of the vector field $F(x, y, z) = x^2yz\,i + xy^2z\,j + xyz^2k$ across S by using the Divergence Theorem, where S is the surface of the box enclosed by the planes $x = 0,\ x = a,\ y = 0,\ y = b,$ $z = 0$ and $z = c$, where a, b, c are positive numbers. $= \frac{3}{4} a^2 b^2 c^2$

$$\iiint_E \operatorname{div} F \; dV$$

$$\frac{\partial Q}{\partial y} = 2y$$

$$\frac{\partial Q}{\partial x} = x^2 + y$$